

Cardiac Output (CO)

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Cardiac Output:

- It is the amount of blood pumped by each ventricle in one minute
- It is the product of heart rate (HR) and stroke volume (SV)
 - HR is the number of heart beats per minute
 - SV is the amount of blood pumped out by ventricle with each beat
 - $SV = \text{ejection fraction} \times EDV$
 - **Ejection fraction** = 2/3 of blood contained and ejected by the heart

...cont'd

Cardiac reserve:

- It is the difference between maximal exercise and resting CO
 - CO = 5-6 liters per minute at rest
 - CO = 20-40 liters per minute at maximum exercise
- E.g. 35L-5 L =30L
- Top endurance athletes may have a cardiac reserve seven or eight times their resting CO.
- People with severe heart disease may have little or no cardiac reserve, which limits their ability to carry out even the simple tasks of daily living.

Cardiac Index:

- It is the amount of blood pumped of the left ventricle each minute to the body surface area.

CI = Cardiac output ÷ Body surface area

E.g. 5.5 L/min ÷ 1.7 = 3.23 L/m²

Heart Rate

- The number of heart beats per minute.
- Heart rate is expressed as beats per minute or bpm.
- Heart rate is used by medical professionals to help **diagnose and track medical conditions**.
- It is also used by athletes to get the most **out of training**.
- A person's heart rate when they are at rest, that is lying down but awake, and not having recently exerted themselves is **Resting Heart Rate**.

Heart Rate

- Normal HR: 60 to 100 beats/minute
 - If < 60 beats/minute, bradycardia but normal in athletes
 - If > 100 beats/minute, tachycardia
- Well trained athletes may have a resting heart rate of 60 bpm or lower.
 - Haile G/Silasse – 36
 - Ryan Hall – 29 (marathoner)

Maximum Heart Rate

- The highest number of heart beats individuals should achieved during physical exertion is depends on age.
- The most common formula used is

$$\text{MHR} = 220 - \text{age}$$

- $220 - 43 = 177$ bpm
- As age increases, MHR decreases

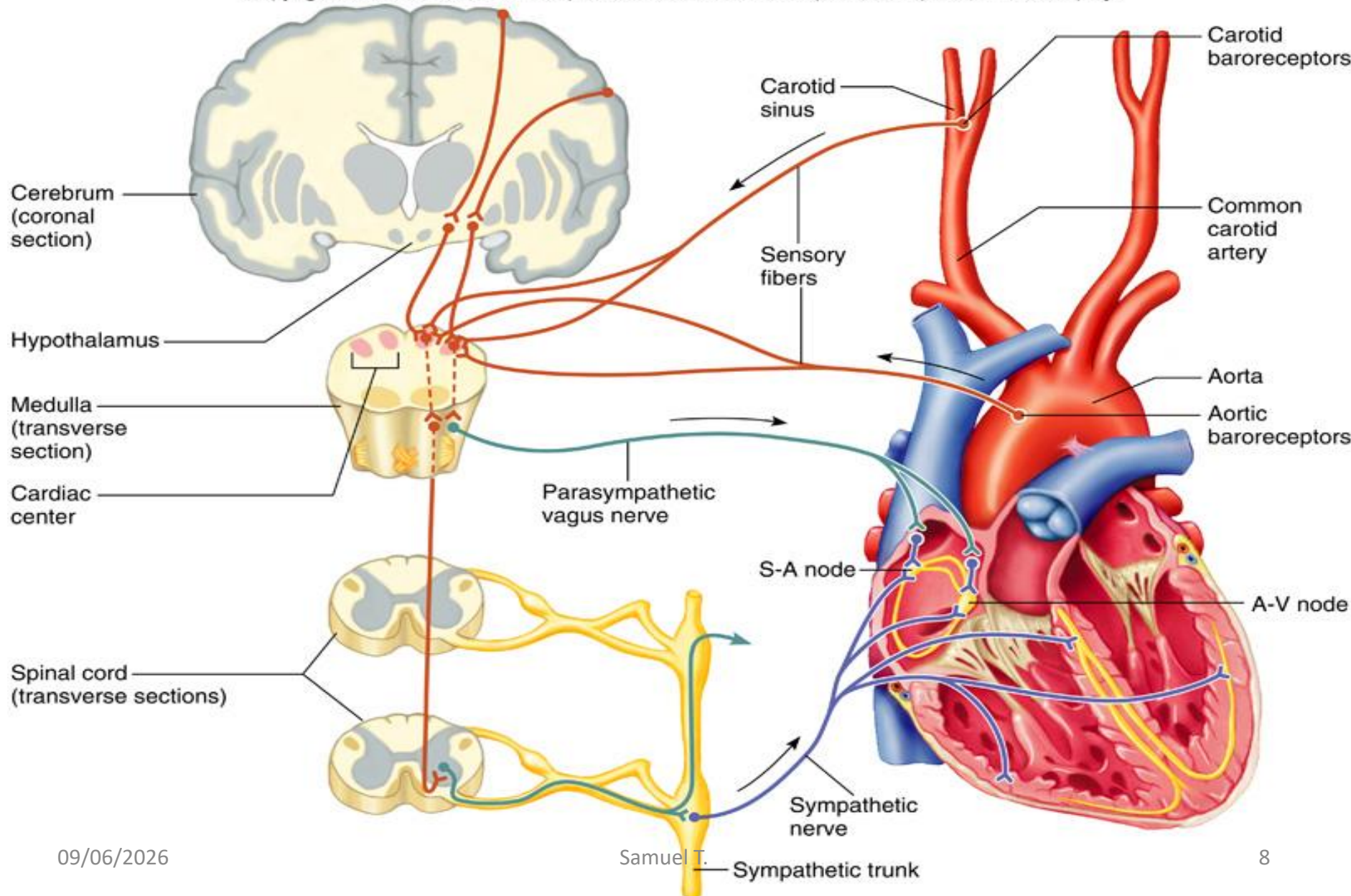
Heart Rate

HR varies with the following factors

1. Age: higher in new born infants (120 b/min)
2. Sex: higher in females (85 b/min)
3. Time of the day: ↓morning, ↑evening
4. Resting and sleep: decreased
5. Physical training: low in athletes (45-60 b/min)
6. Body position: ↑standing, ↓supine positions

Heart Rate Regulation

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Heart Rate Regulation

- **Vagus nerve** (via ACh) ↓ HR by ↓ inflow of Na^+ and Ca^{++} and by ↑ the subsequent outflow of potassium (K^+).
- Acts at SA and AV nodes.
- May treat **SNS-driven heart attack** by gagging or massage of carotid arteries → activate vagal reflexes → PNS counteracts SNS.
- Norepinephrine (NE) and epinephrine (Epi) ↑ inflow of Na^+ and Ca^{++} → increase rate of re-excitation in SA node.
- This ↑ Ca^{++} also increases contractility.
- SNS terminals also excite AV node and whole myocardium: enhances contractility everywhere.

STROKE VOLUME

➤ The amount of blood pumped out of each ventricle per beat is called stroke volume (SV) = 70 ml.

Cardiac Out put= Stroke Volume x Hear Rate

[70 mL x 72 beats / min= 5040 ml. Approx]

❑ Stroke volume is also defined as the difference between the ventricular End Diastolic Volume [EDV] and the End Systolic Volume [ESV].

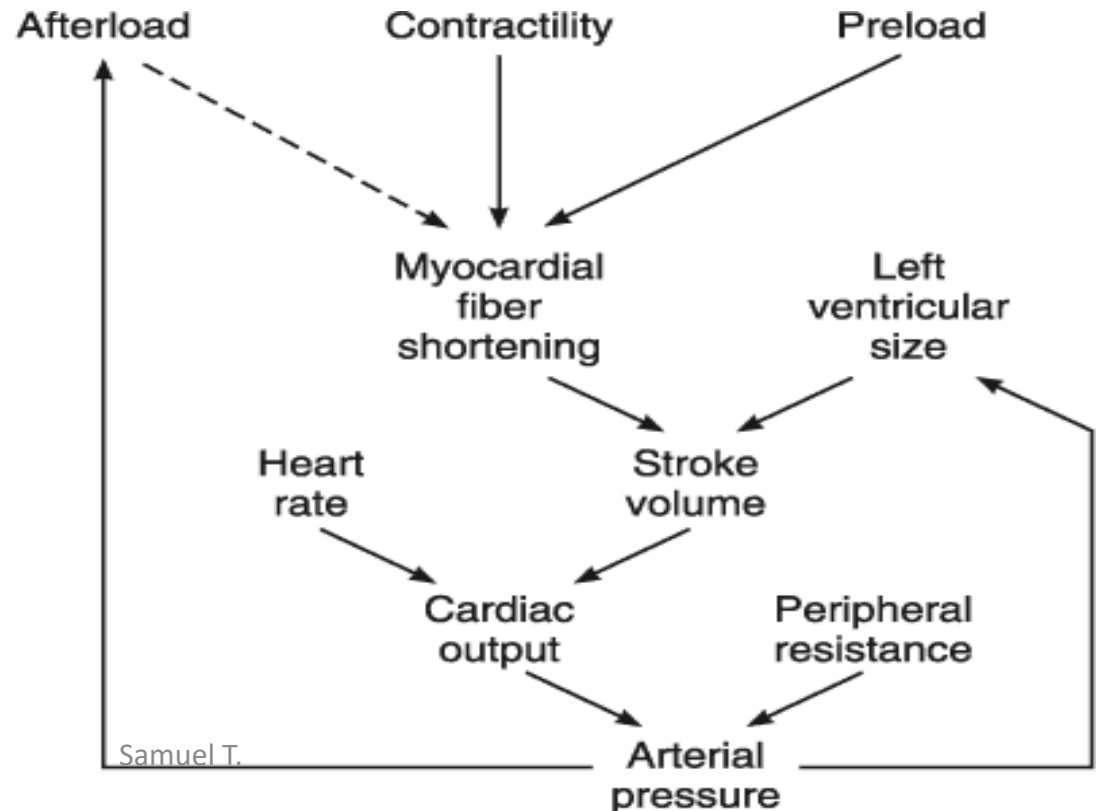
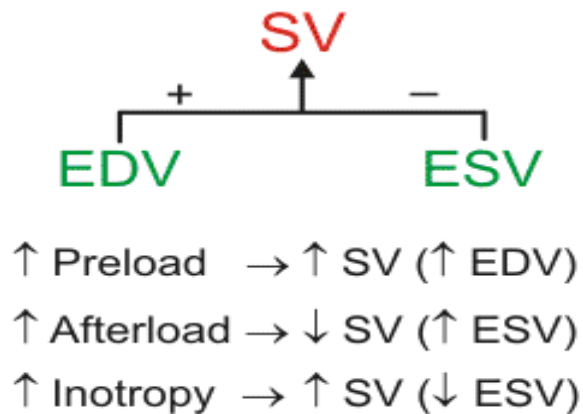
STROKE VOLUME

- ❖ When the heart contracts strongly, the *end-systolic volume* can be decreased to as *little as 10 to 20 milliliters*.
- ❖ Conversely, when large amounts of blood flow into the ventricles during diastole, the ventricular end diastolic volumes can become as *great as 150 to 180 milliliters* in healthy heart.
- ❖ By both increasing the end-diastolic volume and decreasing the end-systolic volume, the stroke volume output can be increased.

Factors Affecting Stroke Volume

❖ Three factors regulate stroke volume

1. Preload /VR=venous return and EDV/.
2. Myocardial contractility/Inotropic agents/.
3. After load



Regulation of Stroke Volume

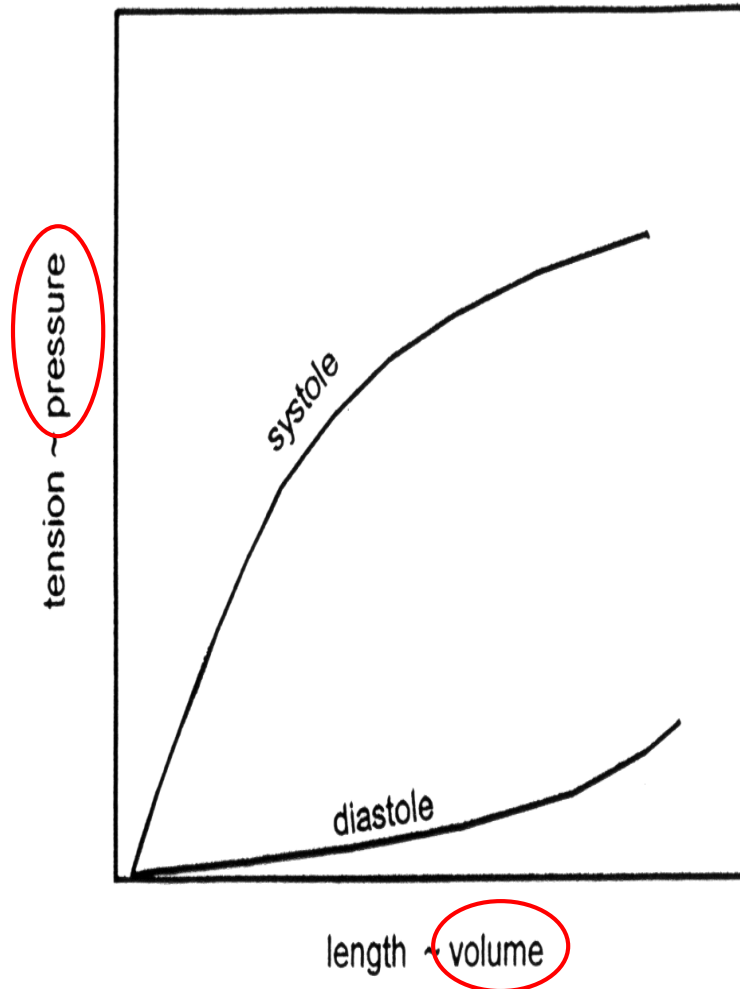
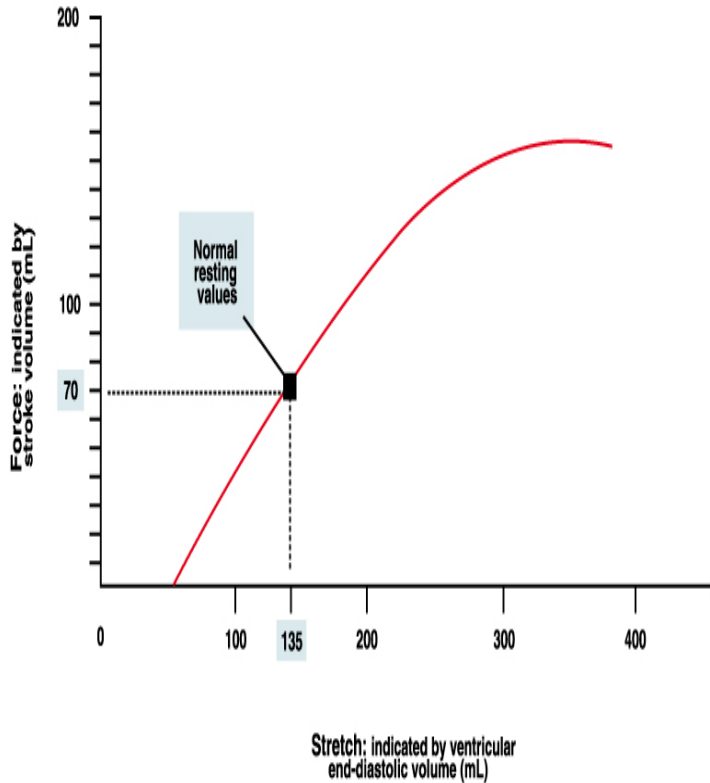
Preload

- A greater preload (stretch) on cardiac muscle fibers prior to contraction increases their force of contraction.
 - *Preload can be compared to the stretching of a rubber band.*
 - *The more the rubber band is stretched, the more forcefully it will snap back.*
- Within limits, the more the heart fills with blood during diastole, the greater the force of contraction during systole.
 - This relationship is known as the **Frank–Starling law** of the heart.

Regulation of Stroke Volume-Preload

- Basically, the Frank-Starling mechanism means that the greater the heart muscle is stretched during filling, the greater is the force of contraction and the greater the quantity of blood pumped into the aorta.
 - *When an extra amount of blood flows into the ventricles, the cardiac muscle itself is stretched to greater length.*
 - *This in turn causes the muscle to contract with increased force because the actin and myosin filaments are brought to a more **nearly optimal degree of overlap** for force generation.*
 - *Therefore, the ventricle, because of its increased pumping, automatically pumps the extra blood into the arteries.*

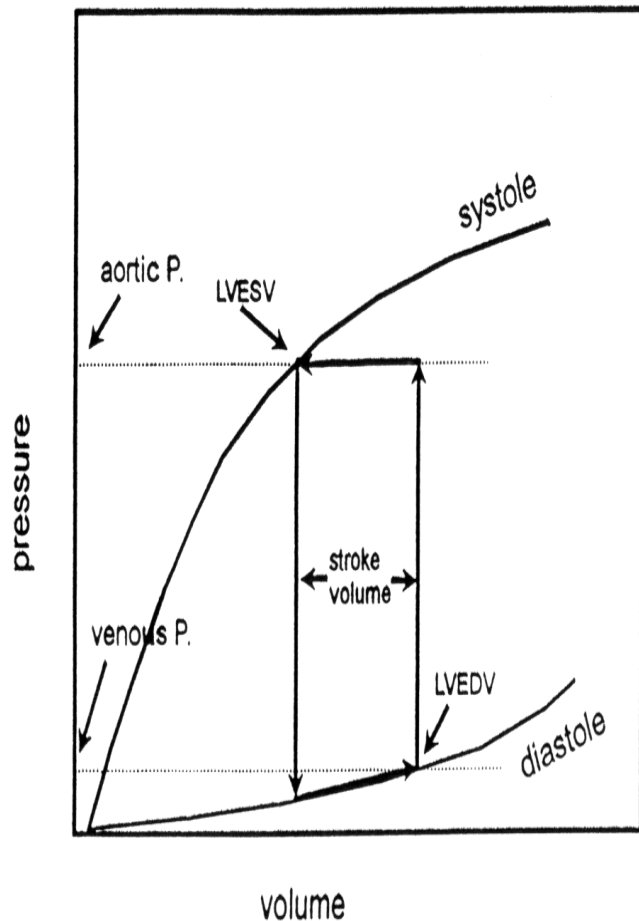
PRESSURE VOLUME LOOP



The length-tension curve for a strip of cardiac muscle can be considered to be equivalent to a pressure-volume curve for a ventricle.

PRESSURE VOLUME LOOP

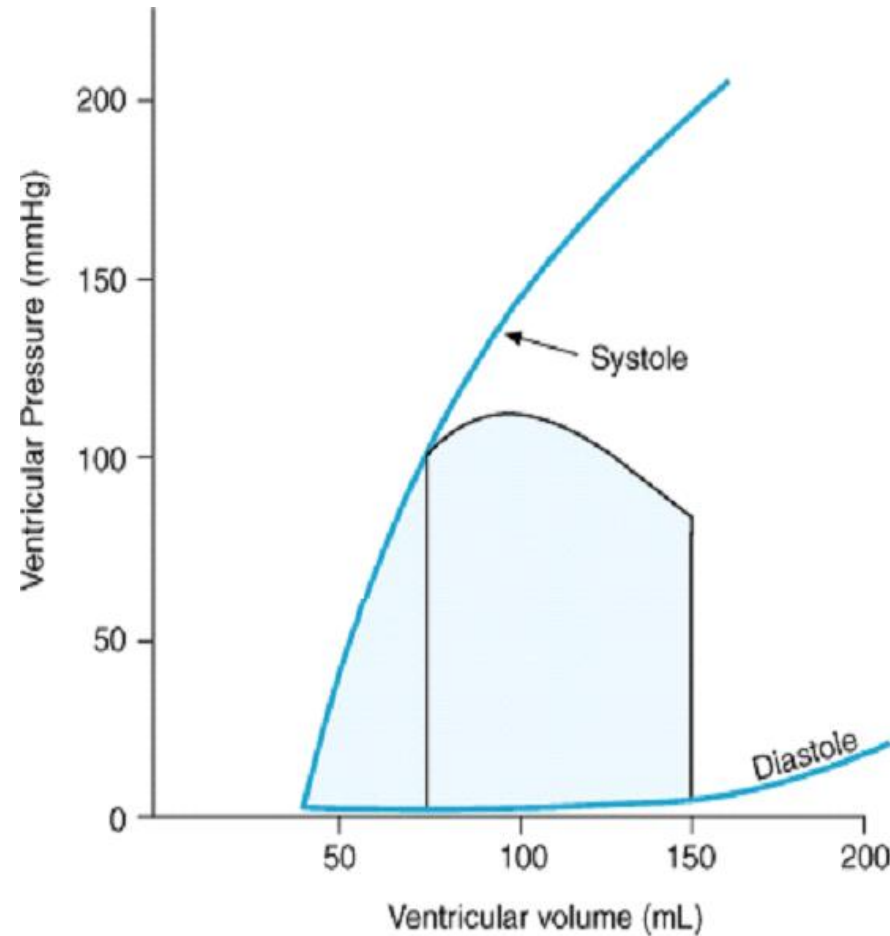
The ejection loop



The heart describes an ejection loop with every beat. The LVEDV is determined by the venous pressure and the diastolic compliance while the LVESV is determined by the aortic pressure and the systolic compliance (contractility).

The stroke volume = $LVEDV - LVESV$.

The ejection fraction = $\text{Stroke Volume} / LVEDV$

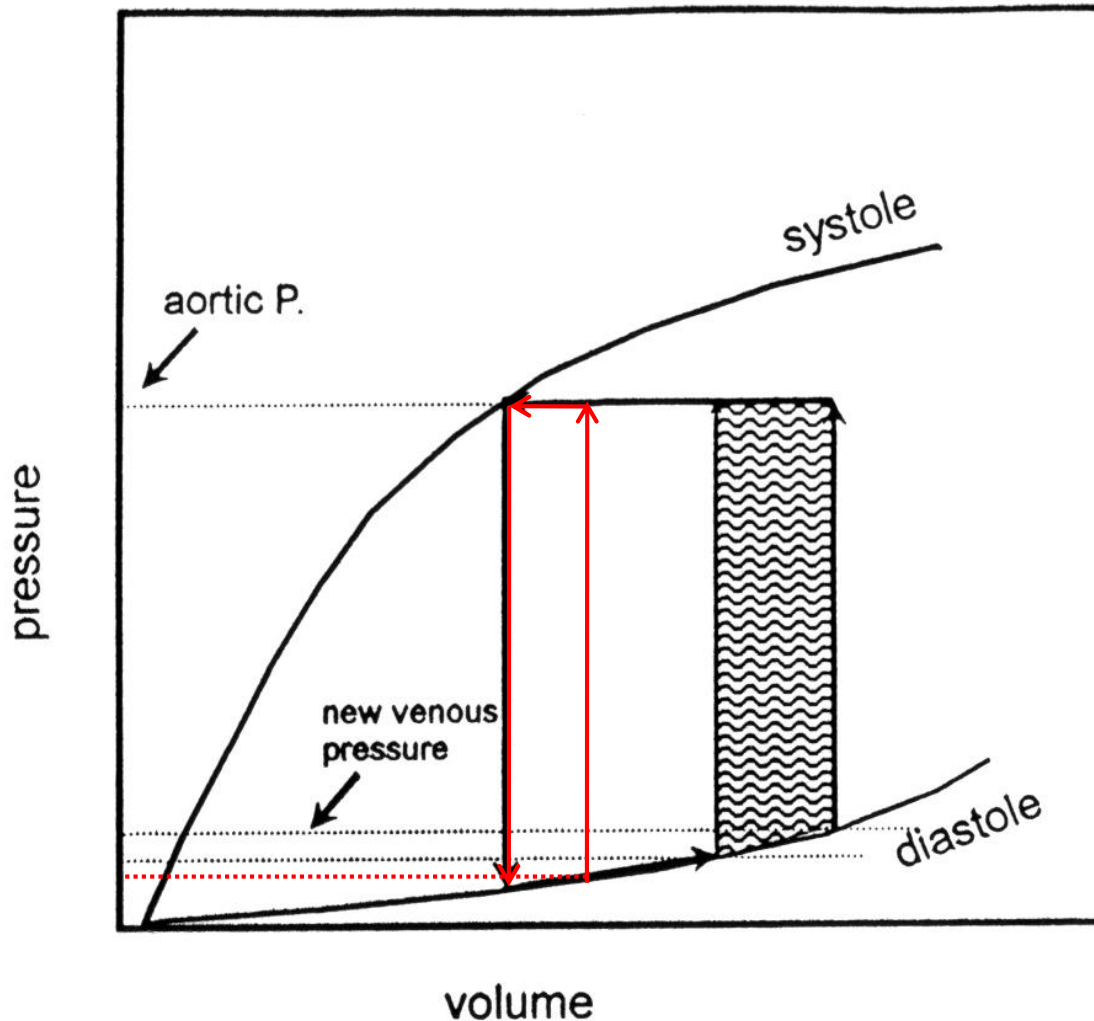


Adapted from Suga and Sagawa, *Circ Res* 1974; 35:117-126.

Regulation of Stroke Volume-Preload

- Two key factors determine EDV:
 1. The duration of ventricular diastole and
 2. Venous return, the volume of blood returning to the right ventricle.
- When heart rate increases (exceeds about 160 beats/min), the duration of diastole is shorter.
 - Less filling time means a smaller EDV, and the ventricles may contract before they are adequately filled.
- By contrast, when venous return increases, a greater volume of blood flows into the ventricles, and the EDV is increased.

PRESSURE VOLUME LOOP - PRELOAD



Increasing venous pressure increases LVEDV and stroke volume but does not affect LVESV.

Changing filling pressure changes stroke volume only by changing LVEDV

This could be an example of transfusion

Lowering LVEDP has the opposite effect (hemorrhage)

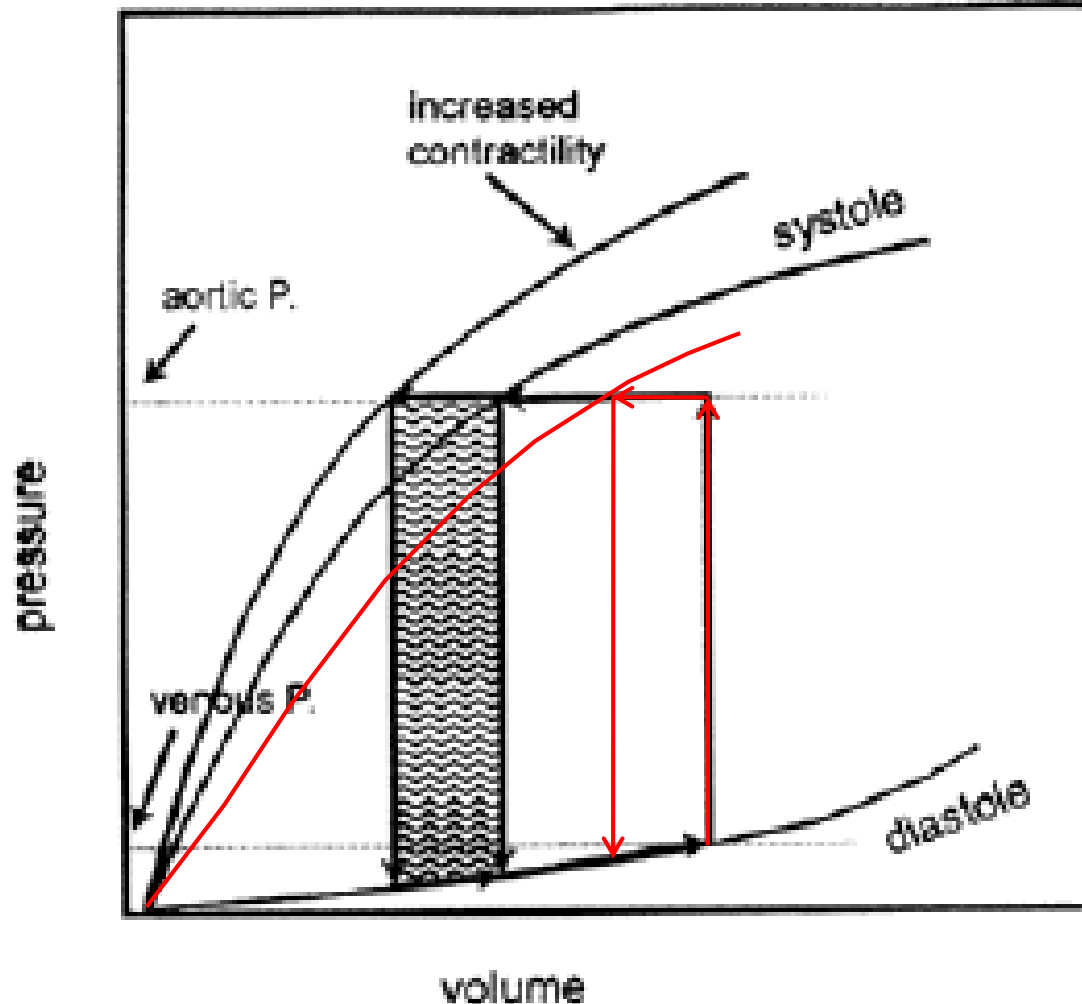
Regulation of Stroke Volume-Contractility

- The second factor that influences stroke volume is myocardial contractility, the *strength of contraction at any given preload*.
- Substances that increase contractility are **positive inotropic agents**; those that decrease contractility are **negative inotropic agents**.
- Thus, for a constant preload, the stroke volume increases when a positive inotropic substance is present.
- Positive inotropic agents often promote **Ca²⁺ inflow** during cardiac action potentials, which strengthens the force of the next contraction.

Regulation of Stroke Volume-Contractility

- Stimulation of the **sympathetic NS**, hormones such as **AD and NAD**, increased **Ca²⁺** level in the interstitial fluid, **and the drug digitalis** all have positive inotropic effects.
- In contrast, inhibition of *SNS*, *anoxia/hypoxia/*, *acidosis*, *some anesthetics*, *and increased K⁺ level in the interstitial fluid* have *negative inotropic effects*.
- Calcium channel blockers are drugs that can have a negative inotropic effect by reducing **Ca²⁺** inflow, thereby decreasing the strength of the heartbeat.

PRESSURE VOLUME LOOP - CONTRACTILITY



Increasing the contractility reduces the LVESV and increases the stroke volume. The LVEDV is not altered.

Increasing contractility decreases LVESV and thus increases stroke volume.

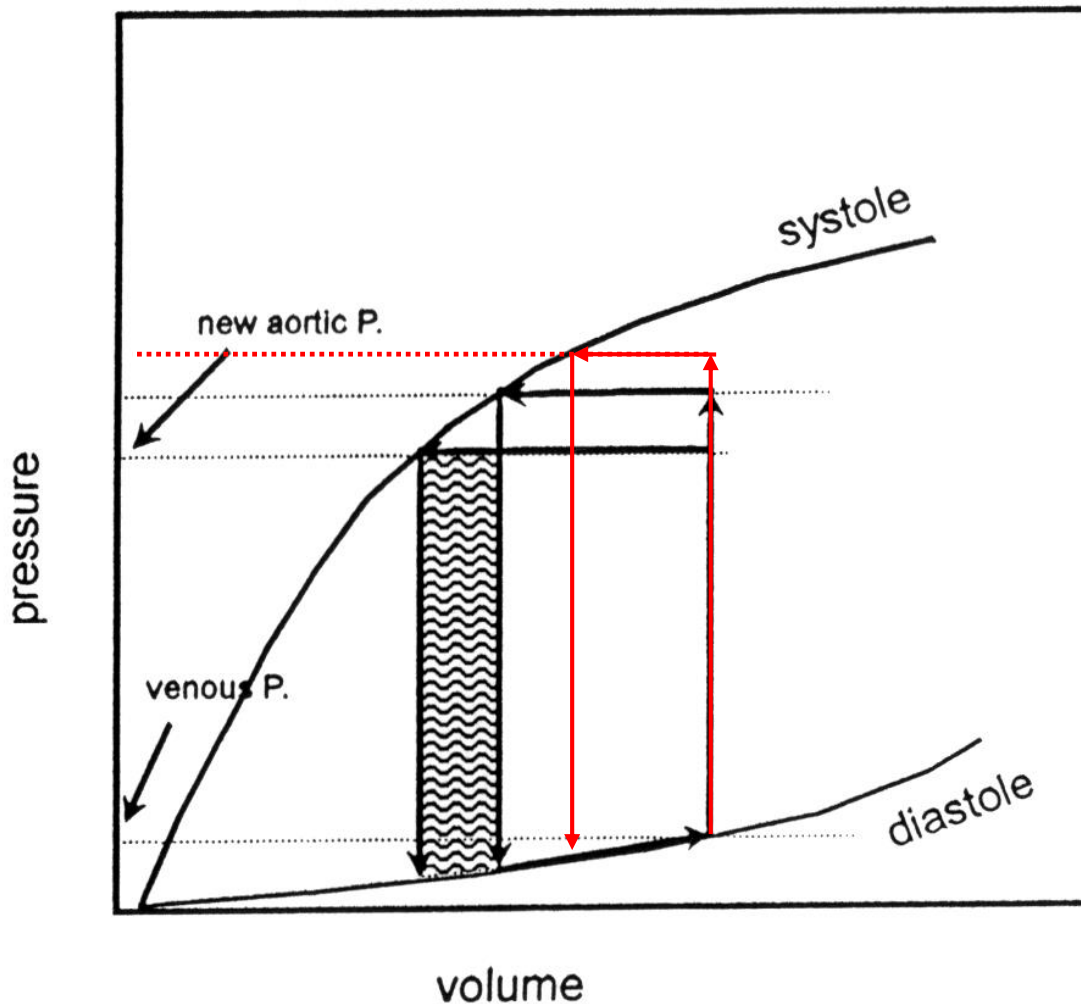
LVEDV is not affected.

Heart failure has the opposite effect

Regulation of Stroke Volume-Afterload

- Ejection of blood from the heart begins when pressure in the ventricle exceeds the pressure in the aorta (about 80 mmHg).
- At that point, the higher pressure in the ventricles causes blood to push the semilunar valves open.
- The afterload of the ventricle is the *pressure in the aorta leading from the ventricle*.
- An increase in afterload causes stroke volume to decrease, so that more blood remains in the ventricles at the end of systole.

PRESSURE VOLUME LOOP - AFTERLOAD



Lowering the aortic pressure decreases LVESV and increases the stroke volume. The LVEDV is unchanged.

Lowering the aortic pressure causes the ventricle to empty more completely.

The stroke volume increases by an amount equal to the fall in LVESV.

LVEDV is not affected.

Raising aortic pressure has the opposite effect

STROKE VOLUME / CARDIAC OUT PUT

Factor

No change Sleep

Moderate changes in environmental temperature

Increase Anxiety and excitement (50–100%)

Eating (30%)

Exercise (up to 700%)

High environmental temperature

Pregnancy

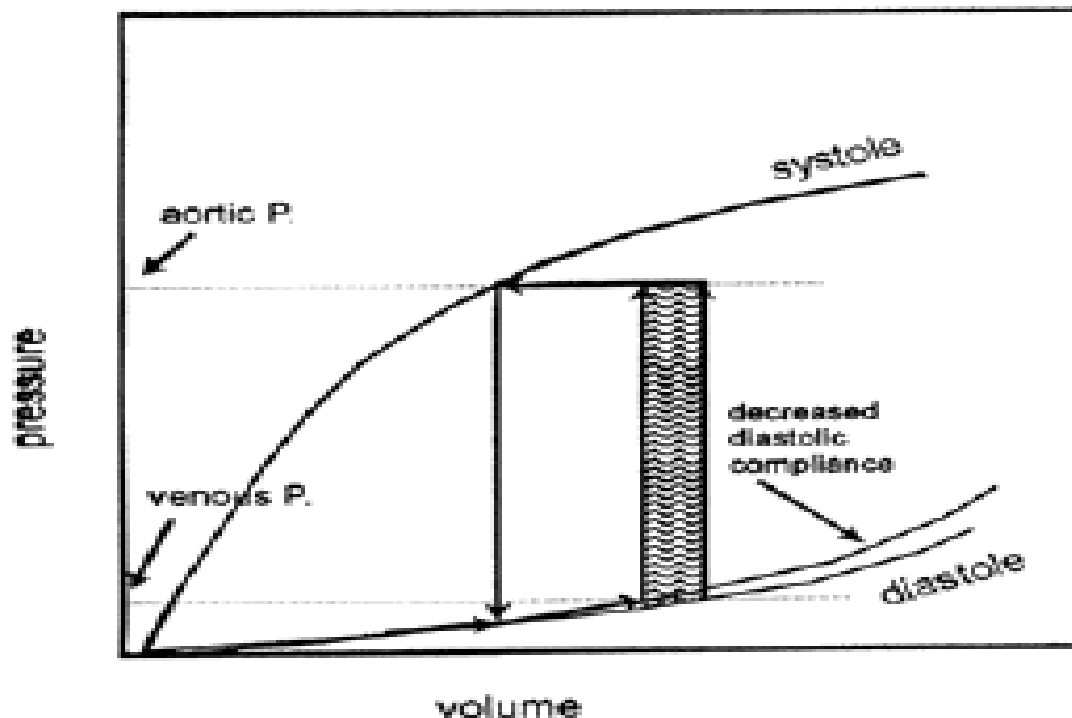
Epinephrine

Decrease Sitting or standing from lying position (20–30%)

Rapid arrhythmias, Heart disease

PRESSURE VOLUME LOOP - COMPLIANCE

- Decreased compliance occurs only in disease and is not a physiological regulator
- *Decreasing the diastolic compliance decreases LVEDV and stroke volume but has no effect on LVESV*



Decreasing the diastolic compliance decreases LVEDV and stroke volume but has no effect on LVESV.

Decreased compliance is a serious problem in patients with hypertrophied hearts as occurs in hypertension.

Venous Return (VR)

Venous return is:

- The volume of blood flowing from veins into RA/min
- 5.5L/min at rest; about 35L/min in well trained athletes during strenuous exercise
- **C.O = VR** at the “operating point” of the cardiac function curves.
- Any factor influencing C.O or VR shifts the “operating point” of C.O/VR intersection to a new position.

Factors affecting Venous Return

1. Mean Systemic Filling Pressure (MSFP)

- Pressure measured every where in the systemic circulation 1 min after blood flow has been stopped by clamping large blood vessels near the heart.
- It is **7 mmHg** and indicates the degree of filling of the systemic circulation
- It is the **driving force** for VR.

2. Right Atrial Pressure (RAP)

- Mean pressure in the right atrium (Central venous pressure) = **2 mmHg**
- Pressure gradient= MSFP-RAP
- $\uparrow \text{RAP} \rightarrow \downarrow \text{VR}$

Factors affectingcont'd

3. Resistance to venous return (RVR)

- Blood meets 1mmHg/L/min resistance during its flow from arterial side to RA
- Occurs mainly at arterioles

$$Q = \Delta P / R$$

- $VR = (MSFP - RAP) / RVR = (MAP - RAP) / TPR$

4. Sympathetic stimulation

- $\uparrow VR$ by inducing vasoconstriction, improved cardiac pumping power.

5. Blood volume

- $\uparrow$ Blood volume $\rightarrow \uparrow VR$ by $\uparrow MSFP$ and vice versa.

6. Respiratory movements

- VR increases with inspiration and decreases with expiration

Factors affectingcont'd

7. Arteriolar dilatation $\rightarrow \downarrow RVR$ and $\uparrow VR$

8. Capillary dilatation $\rightarrow \uparrow$ vascular capacity, $\downarrow MSFP$ and $\downarrow VR$

9. Skeletal muscle contraction

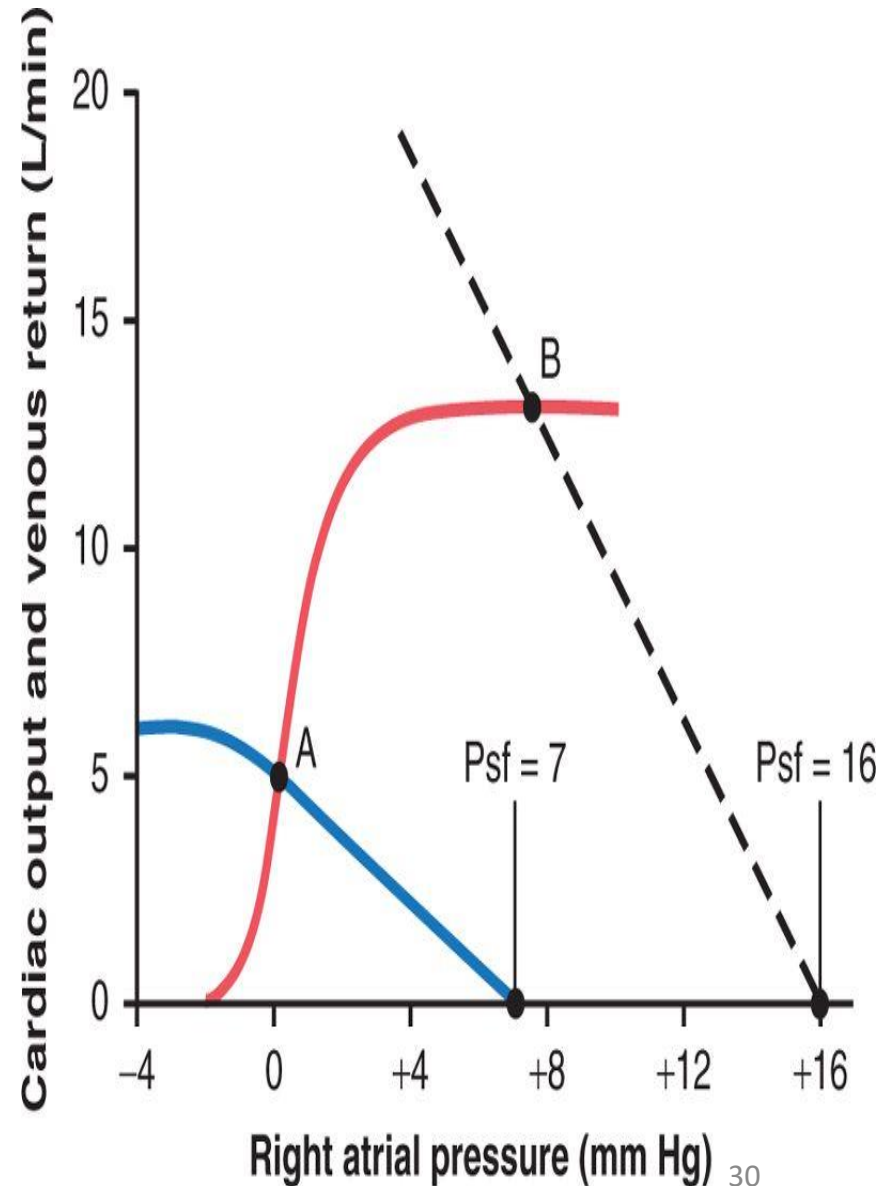
- Squeezes veins b/n muscles $\rightarrow \uparrow MSFP \rightarrow \uparrow VR$

10. Gravity

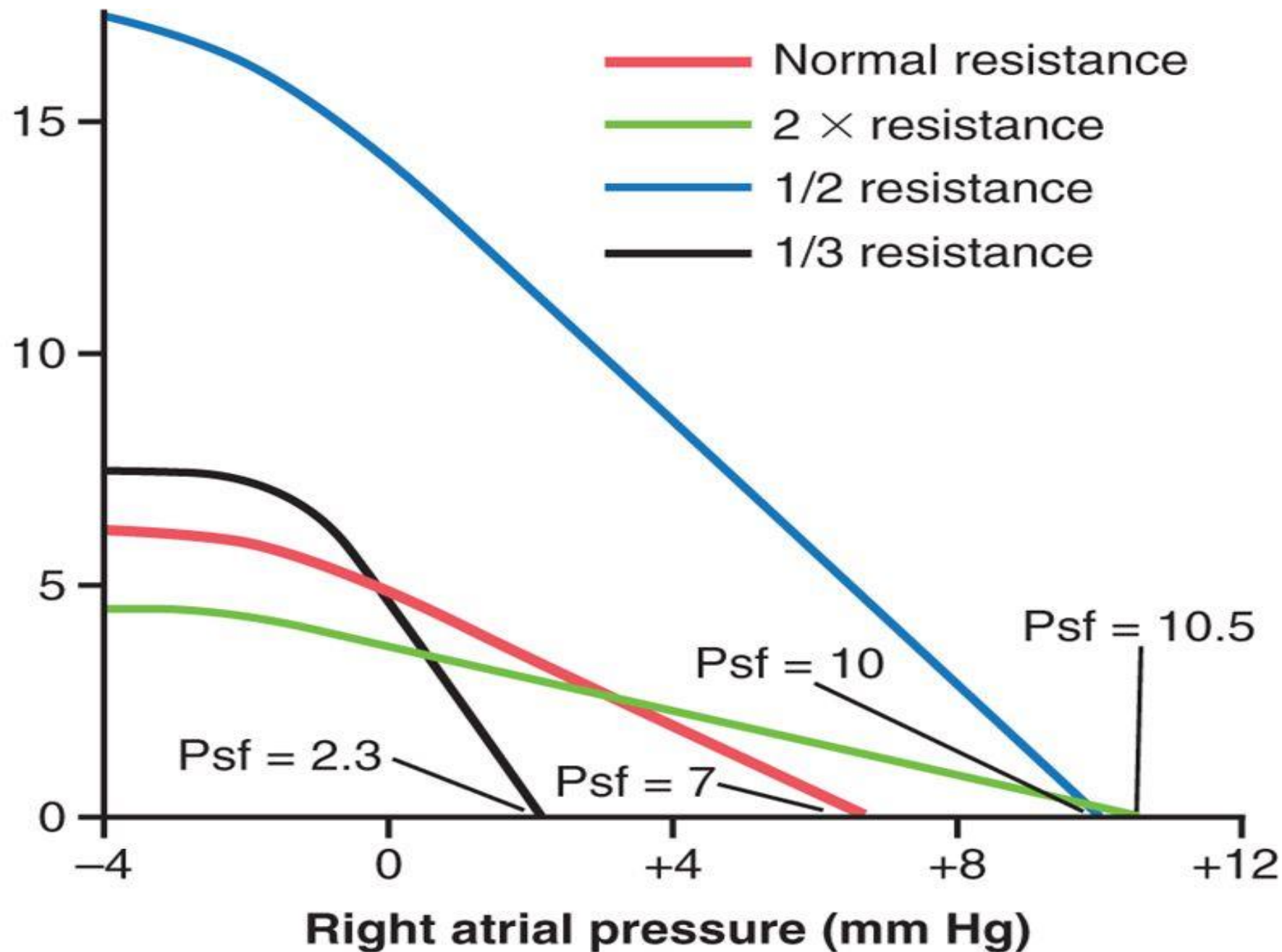
- Standing motionless for some time $\rightarrow$ pooling of blood in lower extremities $\rightarrow \downarrow MSFP \rightarrow \downarrow VR \rightarrow \downarrow C.O \rightarrow$ hypotension $\rightarrow$ brain ischaemia $\rightarrow$ **syncope.**

Cardiac Function Curves

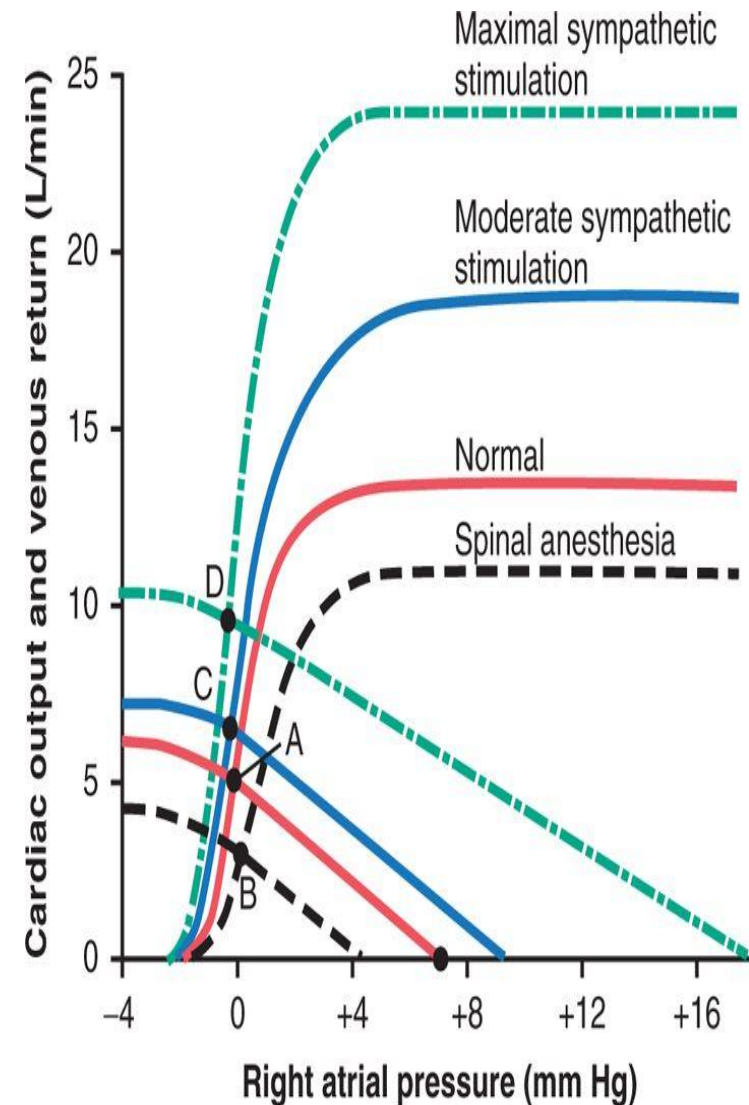
- One can predict the cardiac output and **right atrial pressure** in the following way:
- Determine the **momentary pumping ability** of the heart and depict this in the form of a **cardiac output curve**;
- Determine the **momentary state of flow** from the systemic circulation into the heart and depict this in the form of a **venous return curve**; and
- “**Equate**” these curves against each other.



Venous return (L/min)



- The sympathetic nervous system can be blocked by inducing *total spinal anesthesia*, that blocks transmission of nerve signals through the autonomic ganglia.
- (1) the *mean systemic filling pressure falls to about 4 mm Hg* and
- (2) the *effectiveness of the heart as a pump (SV) decreases to about 80% of normal*.
- The *cardiac output* falls from point A to point B, which is a decrease to about *60 % of normal*.



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Regulation of Pumping Activity of the Heart

- There are two types of regulatory mechanisms:
 1. **The Intrinsic Auto-regulatory Mechanisms (The Frank Starling Mechanism)**
 - ☐ “Within physiological limits, the contractile force of the heart is proportional to the amount of the blood enters the heart”.
 2. **The Extrinsic Regulatory Mechanisms**
 - ☐ Sympathetic, parasympathetic stimulation
 - ☐ Catecholamine
 - ☐ Electrolytes (Ca^{2+} , K^{+})
 - ☐ Drugs: Cardiac glycosides, xanthenes
 - ☐ Hormones: AD, Glucagon, T3/T4

Intrinsic Control of the Heart

- Intrinsic control factors mostly affect **cardiac output** by varying the stroke volume with affecting the **myocardial contractility**
- This can be either heterometric (changes in the sarcomere length) or homometric (independent of the cell length and is controlled by agents such as Na, Ca, pH)

Heterometric Intrinsic Control

- **Frank starling mechanism:** intrinsic control of stroke volume- increase in stroke volume following increase in central venous pressure.
 - *Denervated isolated frog heart, with ligated aorta, was pumped with blood using artificial lung apparatus, warm oxygenated blood from a venous reservoir, height of the reservoir controlled the central venous pressure.*
- ❖ An increase in preload lead to increase in cardiac output .

➤ Factors affecting central venous pressure:

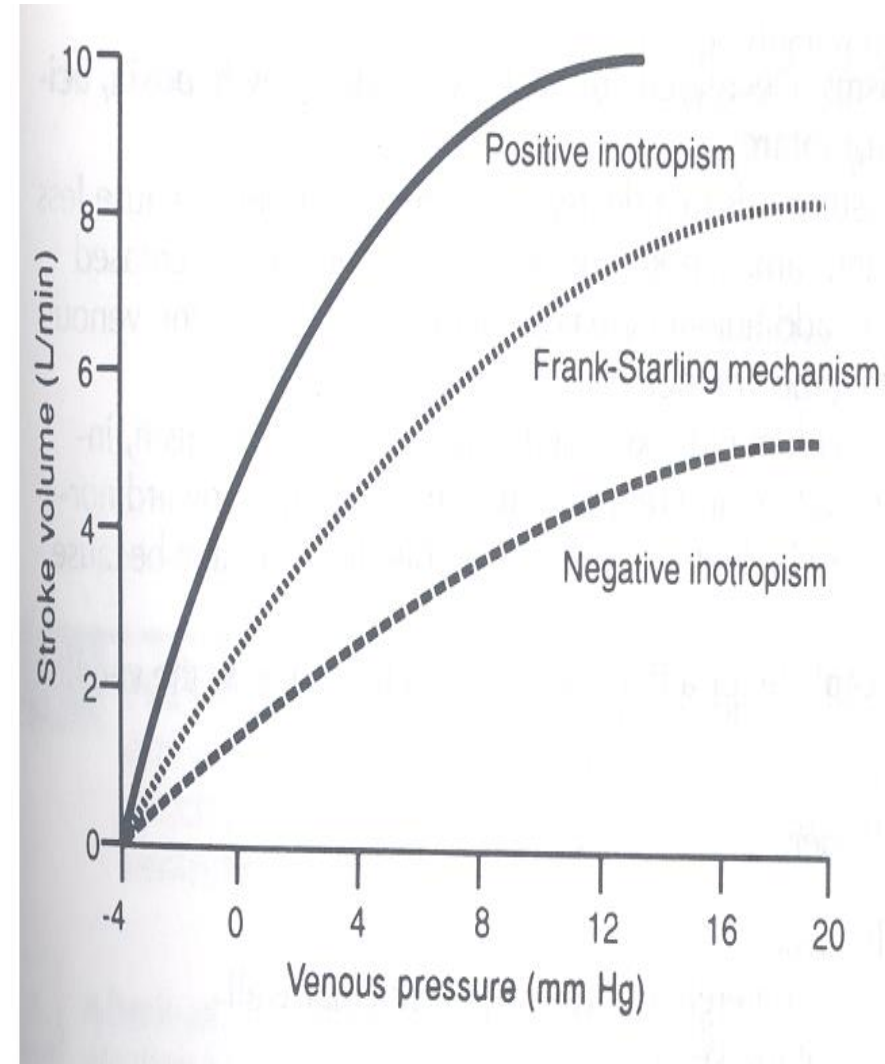
$$\text{Central Venous Pressure} = \text{Blood Volume} / \text{Compliance}$$

- A sudden increase in venous return (VR) → ↑ the heart size.
- For example, *a change from erect to supine position leads to shift of blood from lower extremities to thorax → ↑CVP (4-12cm H₂O) → ↑ in heart size.*

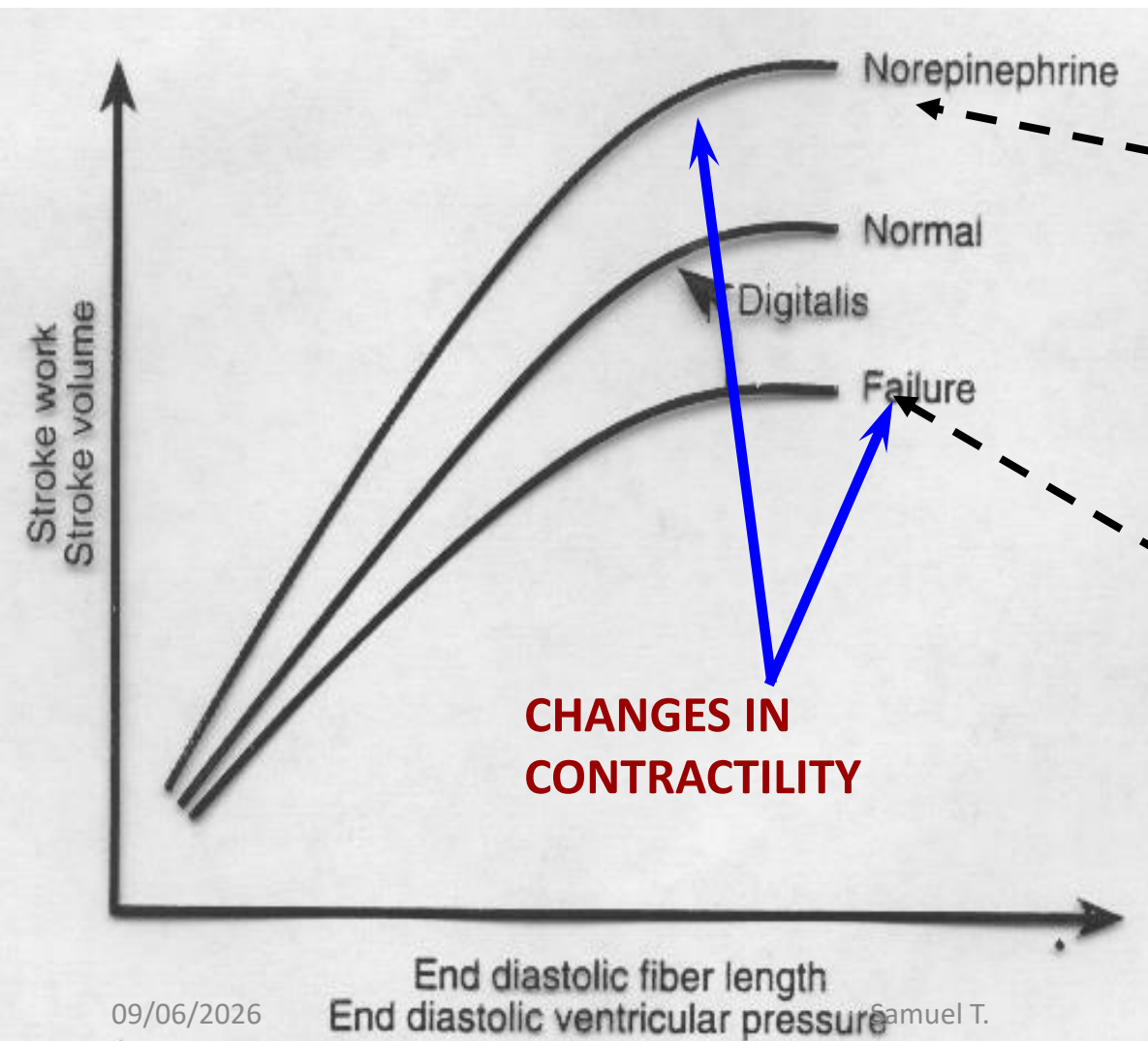
Extrinsic Regulatory Mechanisms

The heart may be:

- **Hyper-effective** (capacity more than normal) due to sympathetic stimulation or hypertrophy.
- **Hypo-effective** and pumping capacity is lower than normal due to:
 - *Sympathetic denervation*
 - *Parasympathetic stimulation*
 - *Atrial flutter or fibrillation*
 - *Myocarditis*
 - *Cardiac hypoxia*



CONTRACTILITY: THE VENTRICULAR FUNCTION CURVE



+ve inotropic agents

- Sympathetic stimulation
- Hypercalcemia
- Glucagon, T3/T4
- Digitalis, Xanthenes
- Catecholamines

-ve inotropic agents

- Parasympathetic stimulation
- Hyperkalemia
- Hypocalcemia
- Acidosis
- Toxins
- Heart diseases

Blood Vessels

Functional Classification of Blood Vessels

The **four main types** of blood vessels are arteries, arterioles, capillaries, and veins

1. **Elastic vessels (Windkassel vessels):**

- Example: Aorta, big arteries
- Pressure storing components
- High ability of recoiling

2. **Resistance vessels (Stop Cock Vessels)**

- Example: small arteries and arterioles
- High muscular component
- Develop high resistance
- Regulate blood flow

Blood vessels: Classification...cont'd

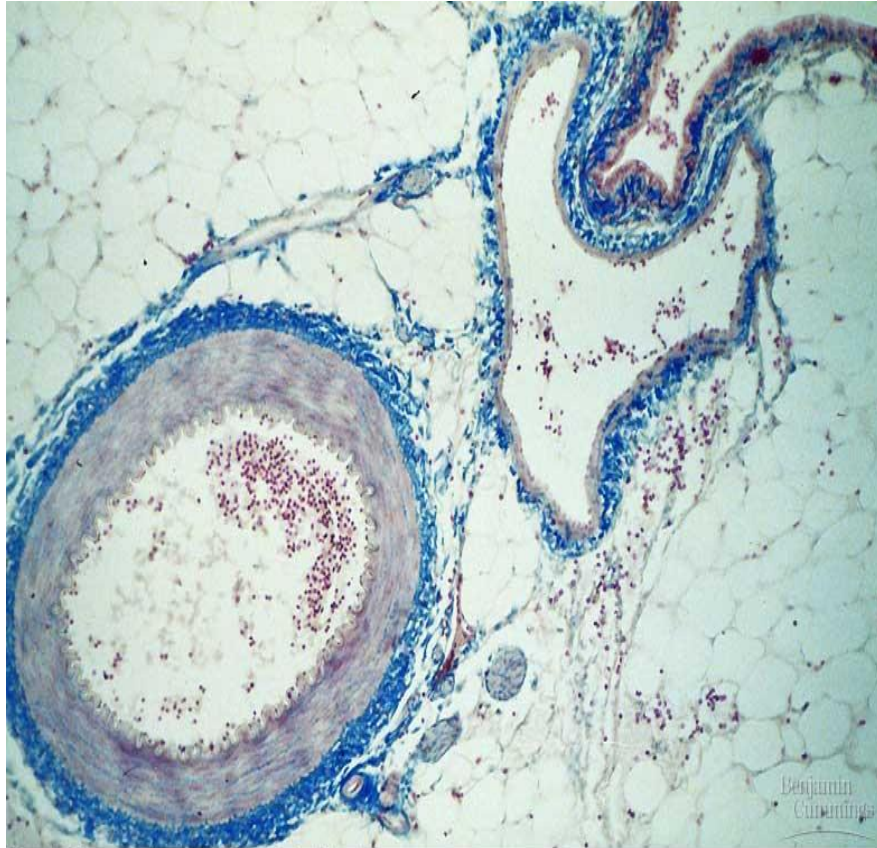
3. Exchange vessels:

- Example: capillaries
- Made up of endothelium and basement membrane
- Thin enough for exchange
- 3 types: continuous, discontinuous and fenestrated capillaries

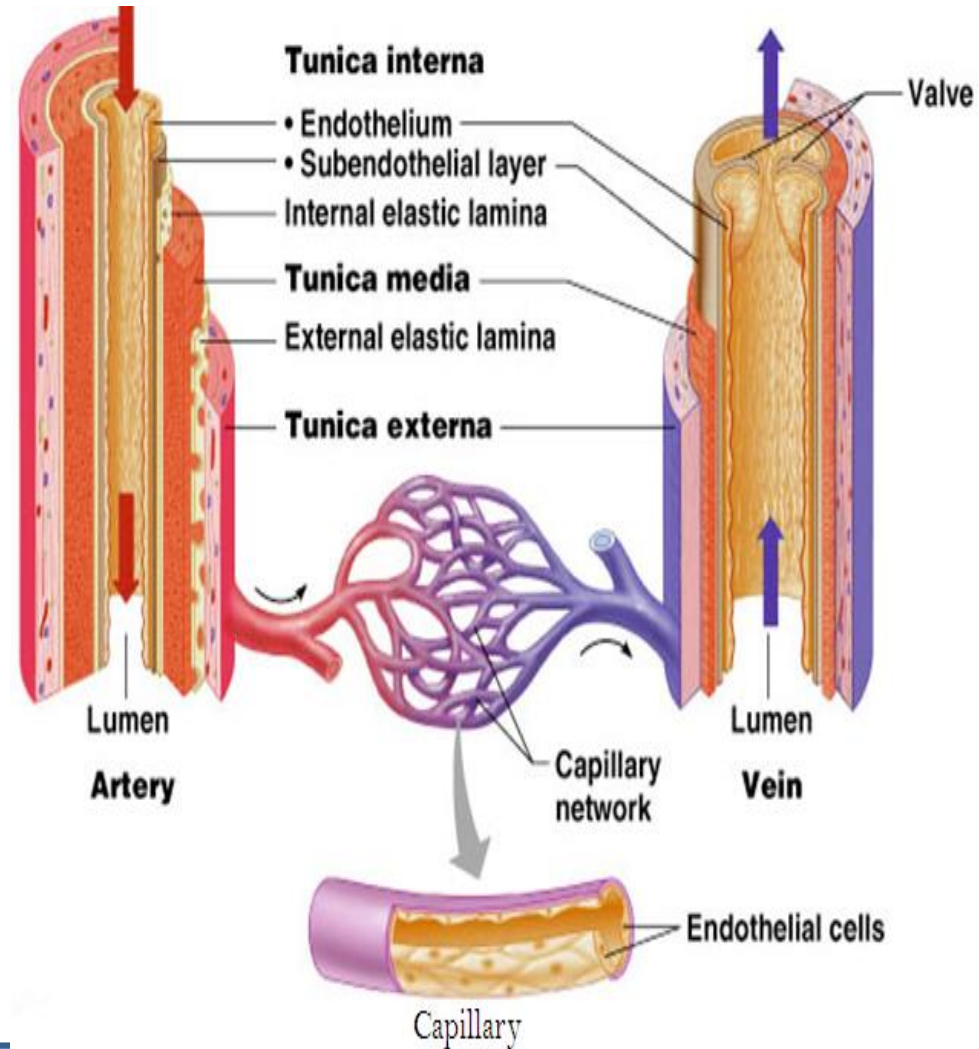
4. Capacitance vessels (big to small veins)

- Very high capacity of distension
- Can accommodate large volume of blood (65% of blood volume)

Histology of blood vessels



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Veins typically contain valves—internal structures that prevent the backflow of blood toward the capillaries.

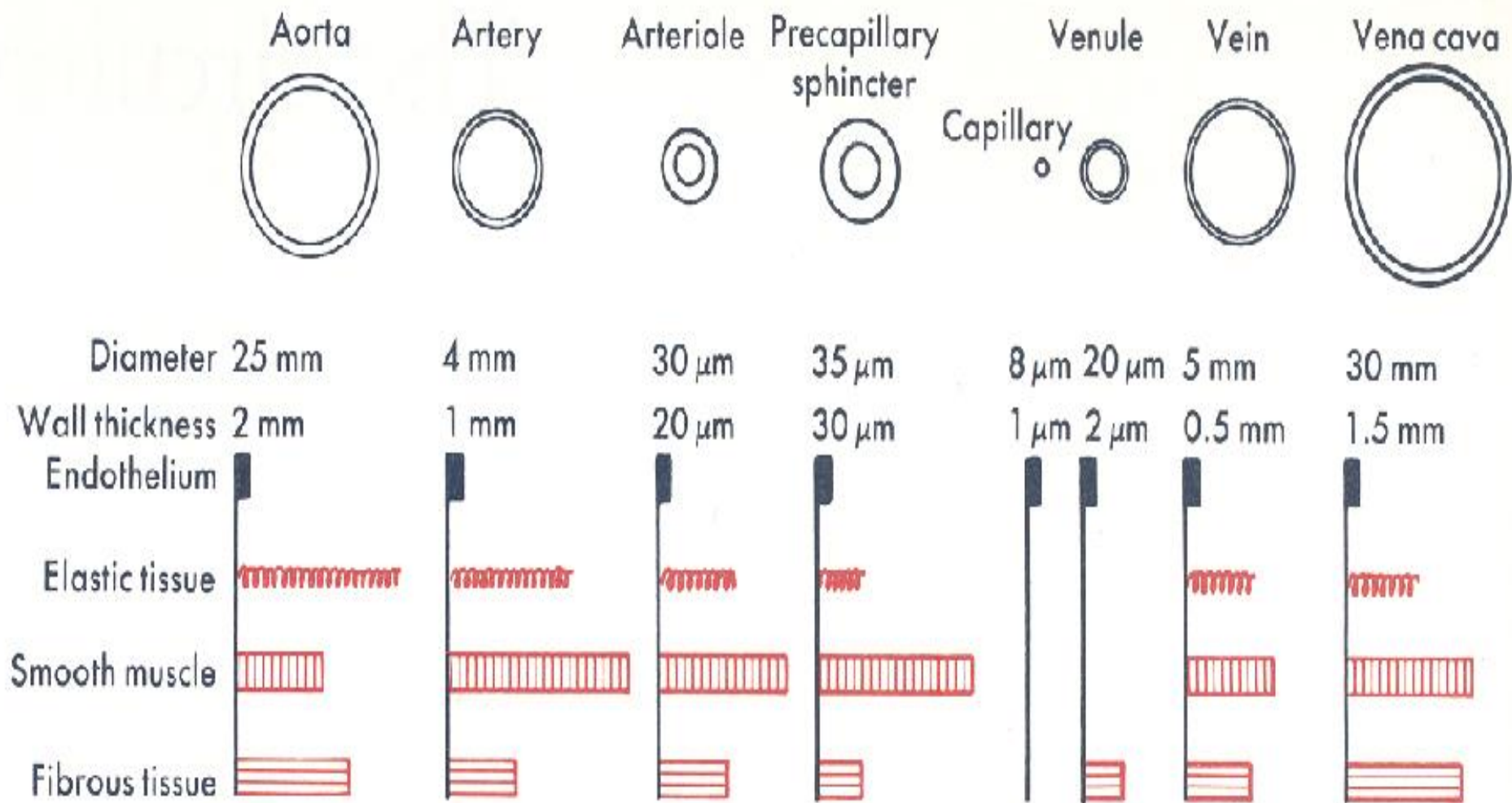
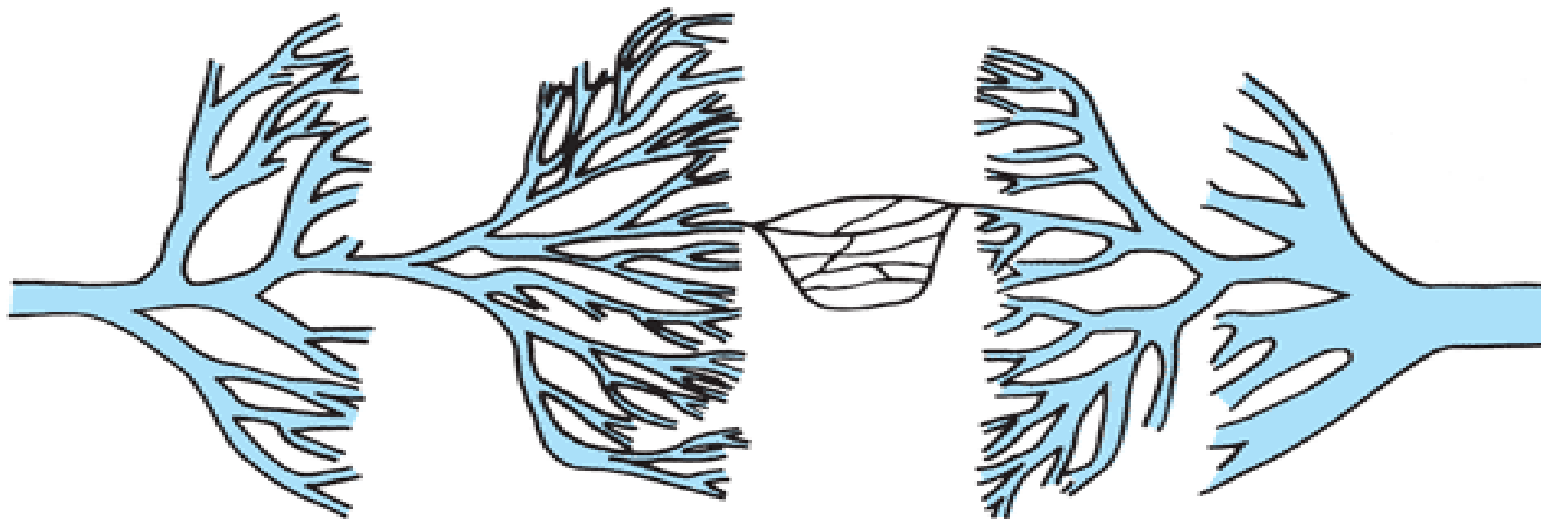


Fig. Internal diameter, wall thickness, and relative amounts of the principal components of the vessel walls of the various blood vessels that compose the circulatory system. Cross sections of the vessels are not drawn to scale because of the huge range from aorta and venae cavae to capillary.



ARTERIES

ARTERIOLES

CAPILLARIES

VENULES

VEINS



Aorta



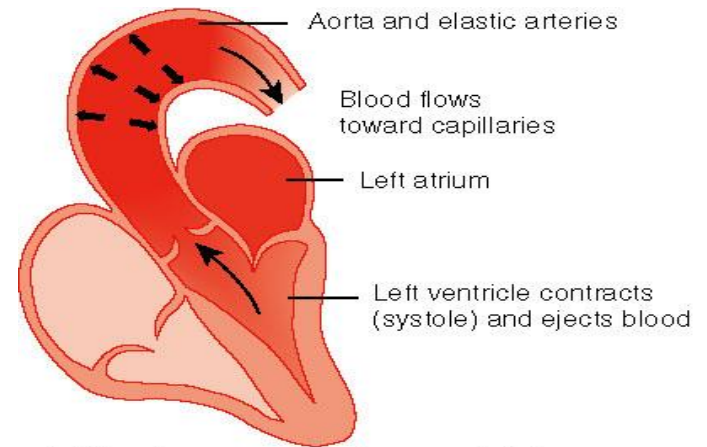
one-way
valves

Venae
cavae

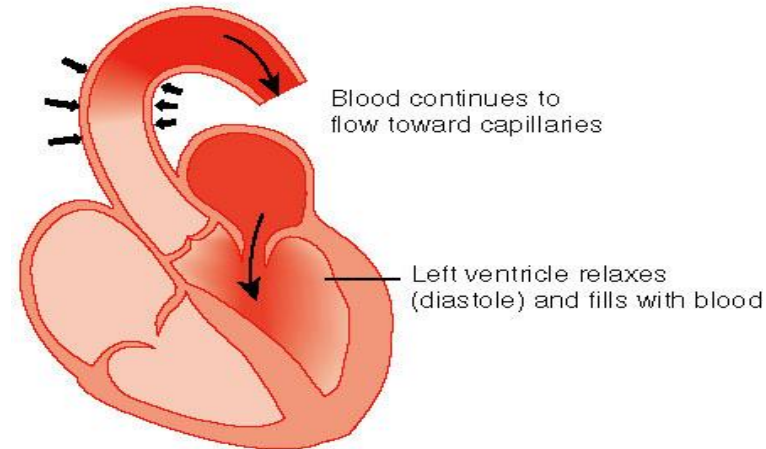
	ARTERIES		ARTERIOLES	CAPILLARIES	VENULES	VEINS	
internal diameter	2.5 cm	0.4 cm	30 μm	5 μm	70 μm	0.5 cm	3 cm
wall thickness	2 mm	1 mm	20 μm	1 μm	7 μm	0.5 mm	1.5 mm
number	1	160	5×10^7	10^{10}	10^8	200	2
total cross-sectional area	4.5 cm^2	20 cm^2	400 cm^2	4500 cm^2	4000 cm^2	40 cm^2	18 cm^2

Elastic (Conducting) Arteries

- Carry blood away from the heart.
- Thick-walled to withstand hydrostatic pressure of the blood during ventricular systole.
- Thick-walled arteries near the heart; the aorta and its major branches
 - ✓ Large lumen allow low-resistance conduction of blood
 - ✓ Contain **elastin in all three tunics**
 - ✓ Withstand and smooth out large blood pressure fluctuations through the body.



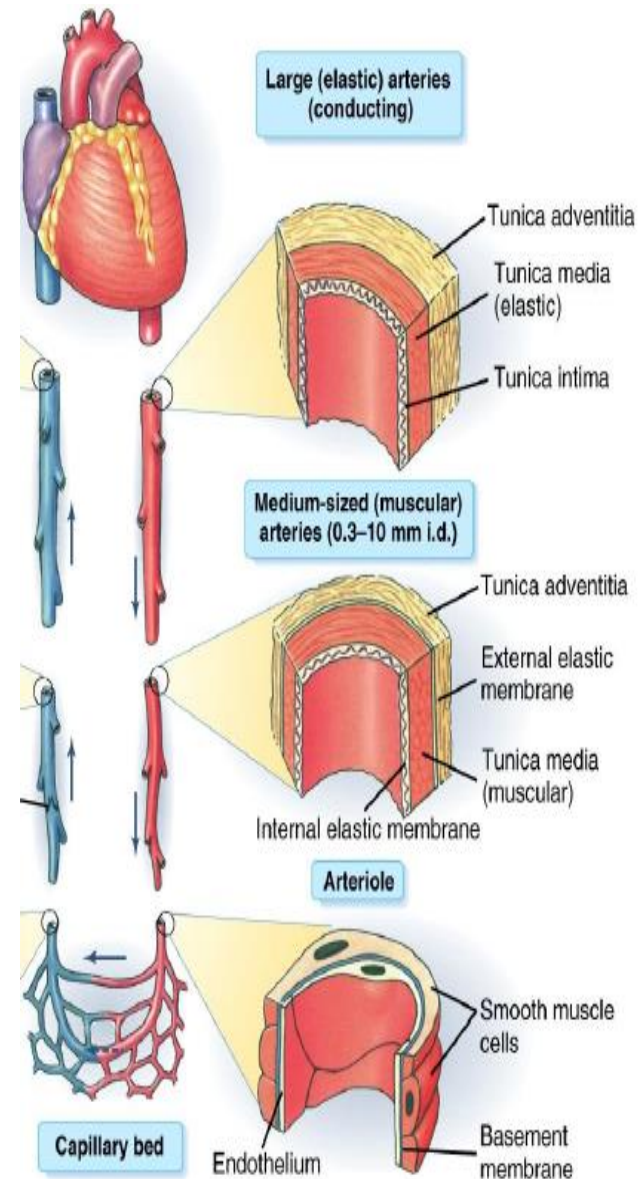
(a) Elastic aorta and arteries stretch during ventricular contraction



(b) Elastic aorta and arteries recoil during ventricular relaxation

Muscular (Distributing) Arteries and Arterioles

- Muscular arteries – distal to elastic arteries; deliver blood to body organs
 - Have thick tunica media with more smooth muscle and less elastic tissue
 - Active in vasoconstriction
- Arterioles – smallest arteries; lead to capillary beds
- This layer is the driving force of the blood vessels by dilating and constricting the diameter of the blood vessels, it regulates **blood flow** and **blood pressure**.

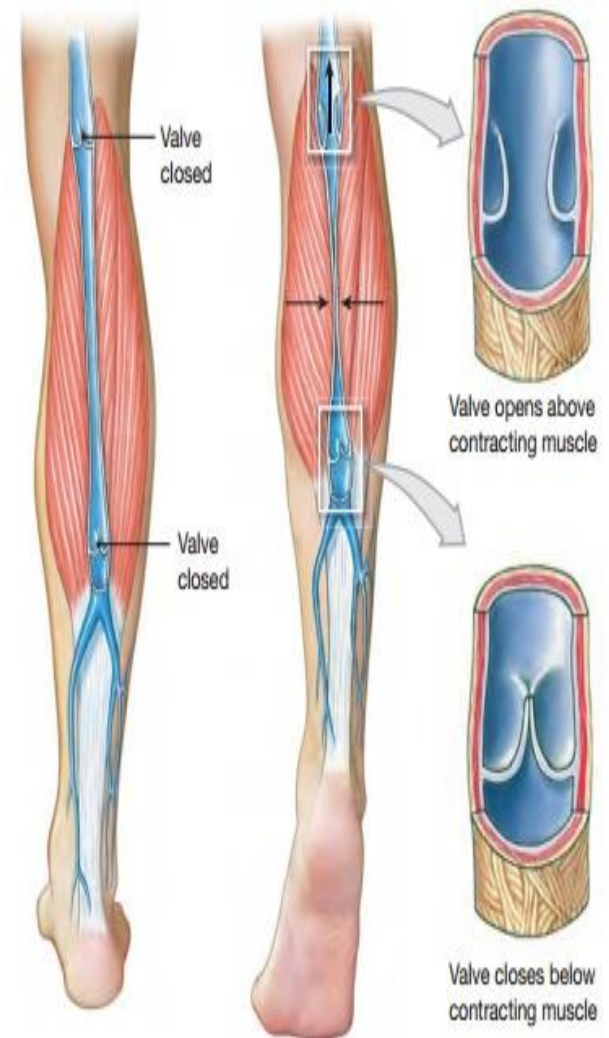


Veins

- Veins collect blood from all tissues and organs and return it to the heart.
- The walls of veins can be thinner than those of corresponding arteries because the blood pressure in veins is lower than that in arteries.
- Venules are the smallest venous vessels that collect blood from capillary beds.
- They vary widely in size and structure.
- The blood pressure in venules and medium-sized veins is so low that cannot overcome easily the force of gravity.

Venous Valves

- Veins of d/t size contain **valves** (folds of the tunica intima that project from the vessel wall and point in the direction of blood flow).
- These valves permit blood flow in one direction only.
- As long as the valves function normally, any movement that **distorts or compresses** a vein pushes blood toward the heart.
- Blood flow due to body movements, not from blood pressure.
- This effect improves venous return, the rate of blood flow to the heart.

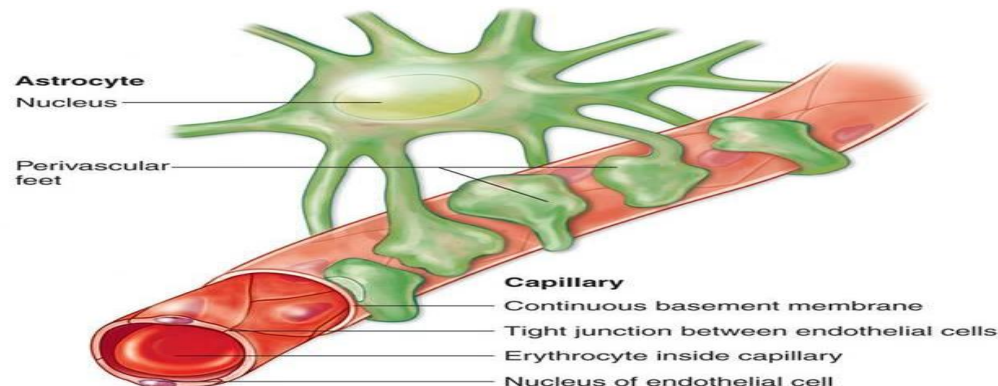


CAPILLARIES

- In the smallest blood vessels (capillaries) the coats are reduced to one, *the tunica intima*.
- Pressure inside is 15 to 35 mmHg, 5% of the blood is in capillaries
- Flow is slow and continuous
- Blood flow controlled by pre-capillary sphincter muscles *via* metabolic regulation
- No smooth muscle, no innervations
- Largest total cross sectional area
- Based on the continuity of their filtration barriers, capillaries have been labelled as **continuous, fenestrated and discontinuous**

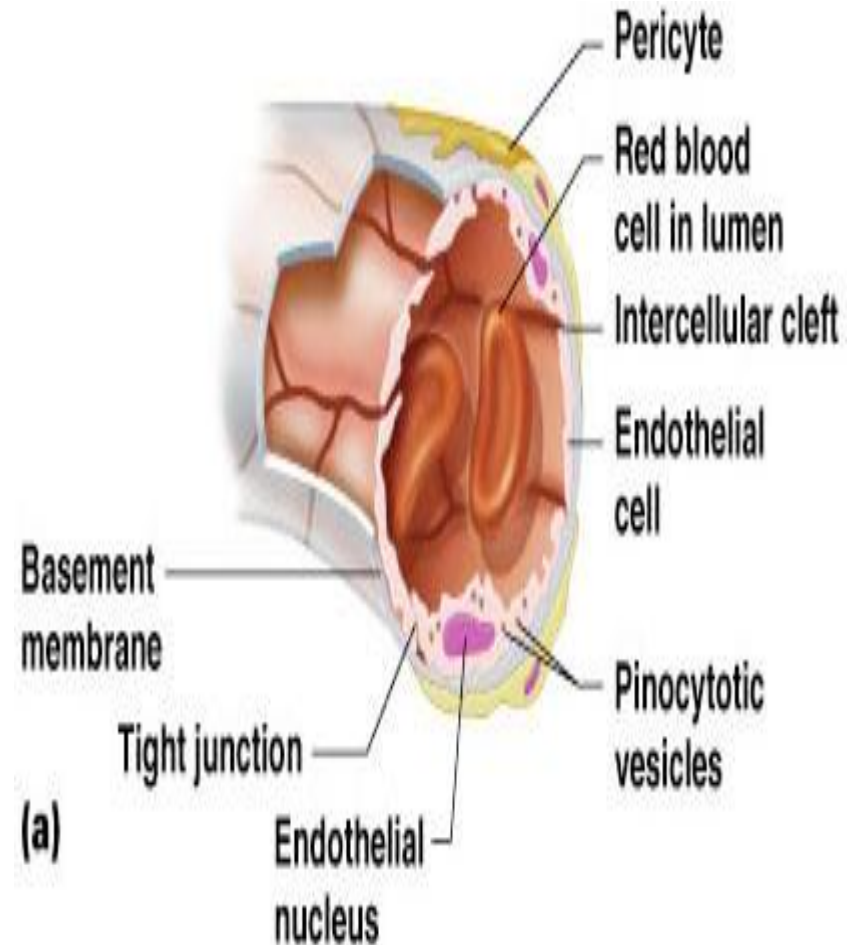
Continuous Capillaries

- ❖ Most regions of the body (skin and muscles) are supplied by continuous capillaries; and have:
 - Endothelial cells that provide an uninterrupted lining
 - Adjacent cells that are held together with **tight junctions**
 - Intercellular clefts of un-joined membranes that allow the passage of fluids
- ❖ Continuous capillaries of the brain:
 - Have tight junctions completely around the endothelium
 - Constitute the blood-brain barrier



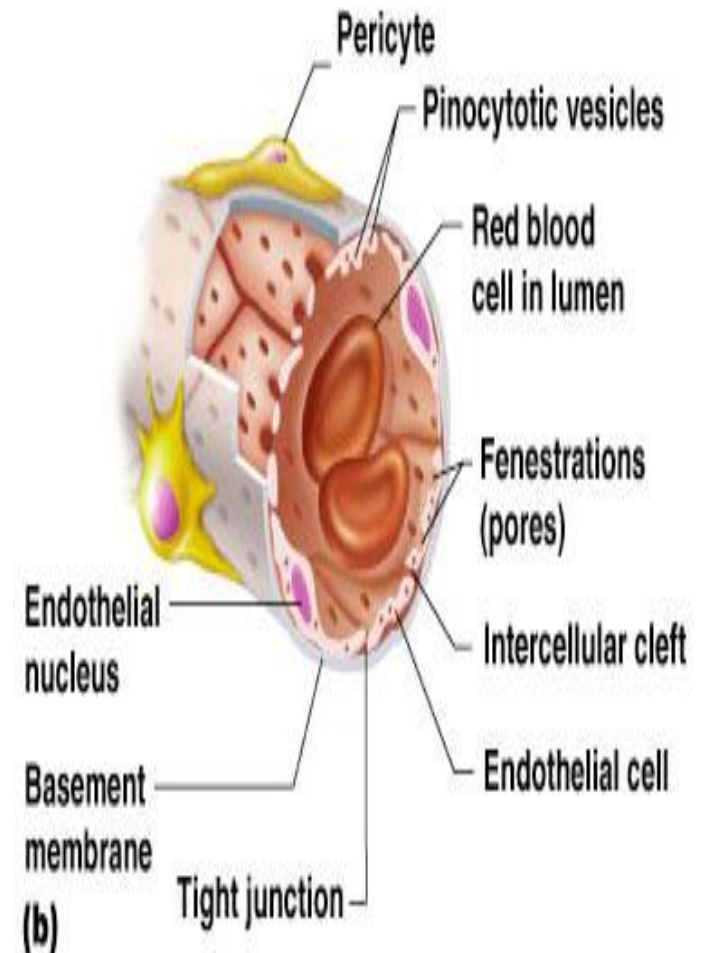
Continuous Capillaries

- Continuous capillaries are located in all tissues *except epithelia and cartilage*.
- Continuous capillaries permit the water, small solutes, and lipid-soluble materials to diffuse into the interstitial fluid.
- At the same time, they prevent the loss of blood cells and plasma proteins.



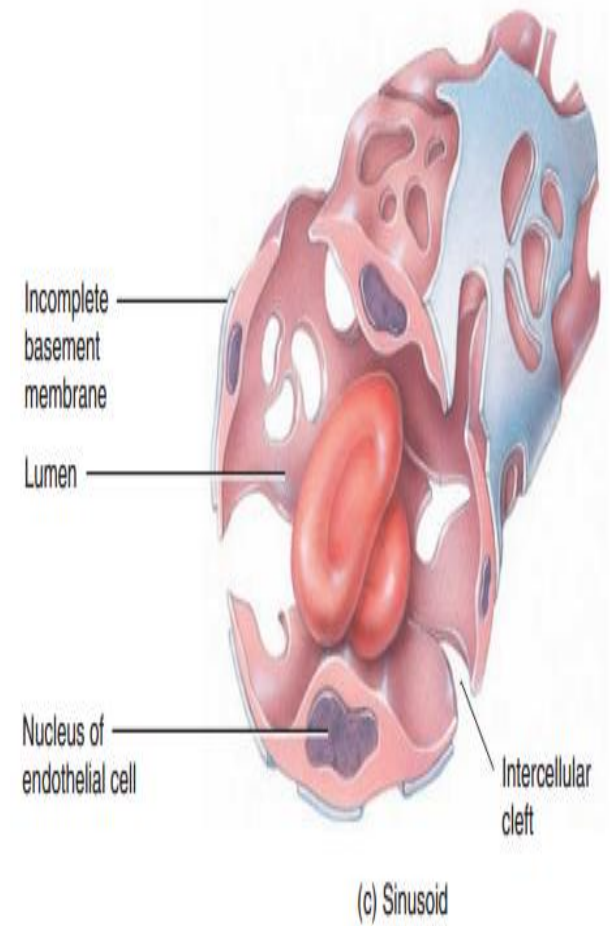
Fenestrated Capillaries

- Other capillaries of the body are **fenestrated capillaries**.
- The plasma membranes of the endothelial cells in these capillaries have many fenestrations, **small pores** (holes) ranging from 70 to 100 nm in diameter.
- Fenestrated capillaries are found in the **kidneys, villi** of the small intestine, choroid plexuses of the ventricles in the brain, **ciliary** processes of the eyes, and in most **endocrine glands**.



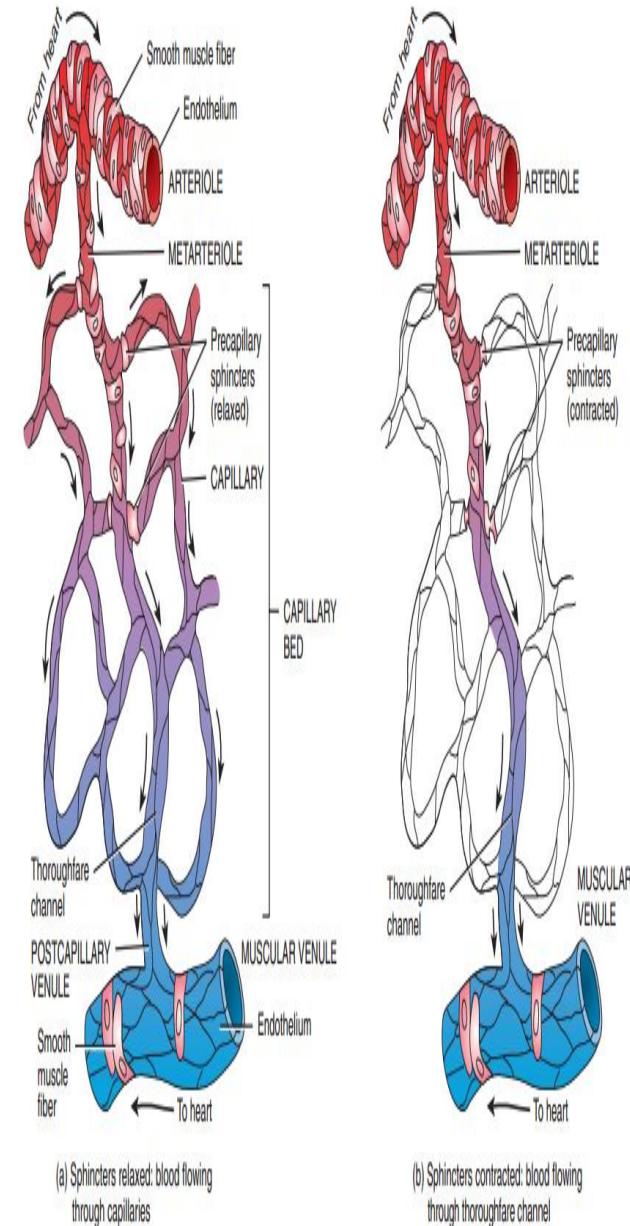
Discontinuous/Sinusoidal Capillaries

- Highly modified, leaky, fenestrated capillaries with large lumens
- Found in the **liver, bone marrow, lymphoid tissue**, and in some **endocrine organs**
- Allow large molecules (proteins and blood cells) to pass between the blood and surrounding tissues
- Blood flows sluggishly, allowing for modification in various ways



CAPILLARIES

- Capillaries form an extensive network, approximately **20 billion interconnecting vessels** that course among the individual cells of the body.
- The flow of blood from a *metarteriole* through capillaries and into a *postcapillary venule* is called the ***microcirculation of the body***.
- The primary function of capillaries is the **exchange of substances** between the blood and interstitial fluid

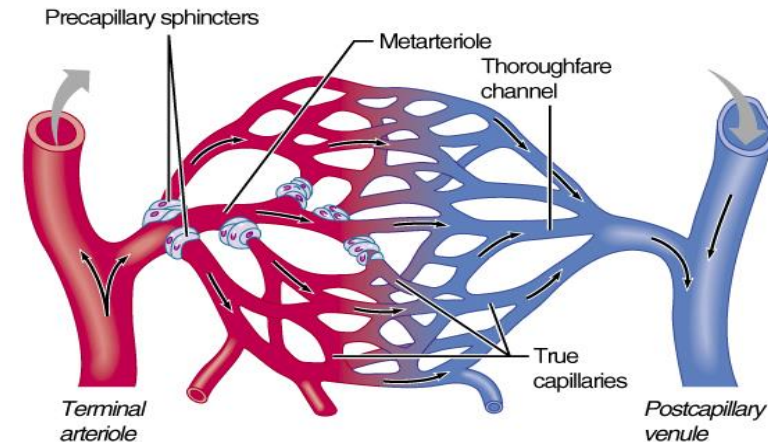


CAPILLARIES

❖ A microcirculation of interwoven networks of capillaries, consisting of:-

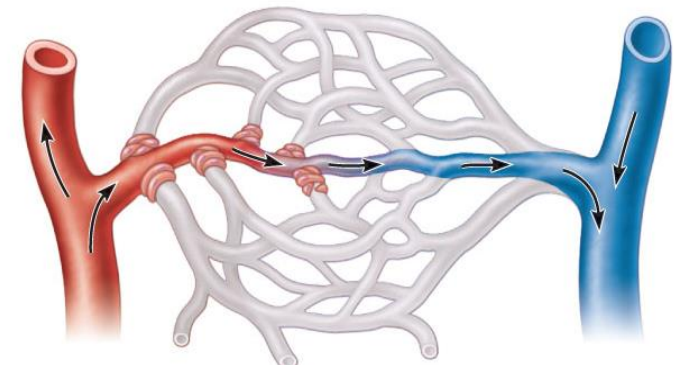
➤ **True capillaries** – 10 to 100 per capillary bed, capillaries branch off the metarteriole and return to the thoroughfare channel at the distal end of the bed.

➤ **Vascular shunts** – metarteriole – thoroughfare channel connecting an arteriole directly with a post-capillary venules



(a) Sphincters open

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(b) Sphincters closed — blood flows through metarteriole thoroughfare channel and bypasses true capillaries.

Blood vessels-Anastomosis

- Most tissues of the body receive blood from more than one artery.
- *The union of branches of two or more arteries supplying the same body region is called an **anastomosis**.*
- Anastomosis between arteries provide alternative routes (collateral circulation) for blood to reach a tissue or organ.
- Anastomosis may also occur between veins and between arterioles and venules.
- Arteries that do not anastomose are known as **end arteries**.
- Obstruction of an end artery interrupts the blood supply to a whole segment of an organ, producing necrosis.

Blood Distribution

- Our total blood volume is **unevenly distributed** among arteries, veins, and capillaries.
- The heart, arteries, and capillaries in the pulmonary and systemic circuits normally contain 30–35% of the blood volume .
- The venous system contains the rest (65–70% = 3.5L).
 - **About one-third** of the blood in the venous system is circulating within the ***liver, bone marrow, and skin.***
 - These organs have extensive venous networks that at any moment contain large volumes of blood.

Blood Distribution

